Abstract linear algebra homework2

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Problem1

(a)

Proof. Since W_1, w_2 are subspaces, which means 0 $\in W_1$,0 $\in W_2$, such that 0 belongs is in the intersection of W_1, W_2. For any two elements w_1, w_2 in the intersection of $W_1 and W_2$,we have following conditions : $w_1, w_2 \in W_1$, since W_1 is a subspace , so $w_1 + w_2$ and cw_1 are all in the W_1 by the definition, subspace W_1 is closure under addition and multiplication. Similarly $w_1, w_2 \in W_2$, $w_1 + w_2$ and cw_1 are also in W_2. Such that $w_1 + w_2$ and $w_1 * w_2$ are all in the intersection of W_1 and W_2. Which means that the intersection of W_1 and W_2 is closure under addition and multiplicaton. To sum up , $W_1 \cap W_2$ is a subspace of V. □

(b)

Proof. Similarily, we hae 0$\in W_1$,0$\in W_2$ so that let w_1=0, w_2=0, so there exist w=$w_1 + w_2$=0 w$\in W_1 + W_2$, such that 0$\in W_1 + W_2$. For addition, letw_1, w_1' is in W_1, w_2, w_2' is in W_2. So for $w + w'$=$(w_1 + w_2) + (w_1' + w_2')$=$(w_1 + w_1') + (w_2 + w_2')$. Since W_1 and W_2 are all subspace, such that $w_1 + w_1'$ is in W_1, similarily for another. Which means it is closure under multiplication. Also for cw=$c(w_1 + w_2)$=$cw_1 + cw_2$ which are also in the subspace. To sum up, the $W_1 + W_2$ is a subspace. □

(c)

Proof. Let $k = \dim(W_1 \cap W_2)$, $m = \dim(W_1)$, $n = \dim(W_2)$.

Step 1: Extend a basis of $W_1 \cap W_2$ to bases of W_1 and W_2.

Choose a basis $S = \{u_1, \ldots, u_k\}$ for $W_1 \cap W_2$. Since $W_1 \cap W_2 \subseteq W_1$, we may extend S to a basis

$$\mathcal{B}_1 = \{u_1, \ldots, u_k, v_1, \ldots, v_{m-k}\}$$

for W_1, and similarly extend S to a basis

$$\mathcal{B}_2 = \{u_1, \ldots, u_k, w_1, \ldots, w_{n-k}\}$$

for W_2. Set $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 = \{u_1, \ldots, u_k, v_1, \ldots, v_{m-k}, w_1, \ldots, w_{n-k}\}$.

Step 2: $\mathcal{B}$ spans $W_1 + W_2$.

Every element of $W_1 + W_2$ has the form $x + y$ with $x \in W_1$, $y \in W_2$. Since $\mathcal{B}_1$ spans W_1 and $\mathcal{B}_2$ spans W_2, $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ spans $W_1 + W_2$.

Step 3: $\mathcal{B}$ is linearly independent.
Suppose

$$\sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{m-k} b_j v_j + \sum_{\ell=1}^{n-k} c_\ell w_\ell = 0. \qquad (*)$$

Rearranging,

$$\sum_{\ell=1}^{n-k} c_\ell w_\ell = -\sum_{i=1}^{k} a_i u_i - \sum_{j=1}^{m-k} b_j v_j \ \in \ W_1.$$

But the left side also lies in W_2 (since each $w_\ell \in W_2$), so $\sum c_\ell w_\ell \in W_1 \cap W_2$. Hence we can write $\sum c_\ell w_\ell = \sum_{i=1}^{k} d_i u_i$ for some scalars d_i. This gives $\sum c_\ell w_\ell - \sum d_i u_i = 0$; by the linear independence of $\mathcal{B}_2$ we conclude $c_1 = \cdots = c_{n-k} = 0$ and $d_1 = \cdots = d_k = 0$. Substituting back into $(*)$ and using the linear independence of $\mathcal{B}_1$ gives $a_1 = \cdots = a_k = b_1 = \cdots = b_{m-k} = 0$. Therefore $\mathcal{B}$ is linearly independent.

Conclusion.
$\mathcal{B}$ is a basis for $W_1 + W_2$ of size $k + (m-k) + (n-k) = m + n - k$, so

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2). \qquad \square$$

Problem 2

If $V = W_1 \oplus W_2$ (i.e. $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$), then $\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2)$.

Proof. Since $W_1 \cap W_2 = \{0\}$, we have $\dim(W_1 \cap W_2) = 0$. Applying the dimension formula from Problem 1(c) to $V = W_1 + W_2$:

$\dim(V) = \dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) = \dim(W_1) + \dim(W_2) - 0 =$
$\dim(W_1) + \dim(W_2)$. $\square$

Problem 3

Let $(v_1, \ldots, v_n)$ be a linearly independent list in V, and let $w \in V$. Then $(v_1, \ldots, v_n, w)$ is linearly *dependent* if and only if $w \in \operatorname{span}(v_1, \ldots, v_n)$.

Proof. ($\Rightarrow$) Suppose $(v_1, \ldots, v_n, w)$ is linearly dependent. Then there exist scalars $a_1, \ldots, a_n, c \in F$, not all zero, such that

$$a_1 v_1 + \cdots + a_n v_n + c\, w = 0.$$

We claim $c \neq 0$. Indeed, if $c = 0$, then $a_1 v_1 + \cdots + a_n v_n = 0$ with not all $a_i = 0$, contradicting the linear independence of $(v_1, \ldots, v_n)$. Therefore $c \neq 0$, and we may solve:

$$w = -\frac{a_1}{c} v_1 - \cdots - \frac{a_n}{c} v_n \ \in \ \operatorname{span}(v_1, \ldots, v_n).$$

($\Leftarrow$) Suppose $w \in \operatorname{span}(v_1, \ldots, v_n)$. Then $w = a_1 v_1 + \cdots + a_n v_n$ for some $a_i \in F$, so

$$a_1 v_1 + \cdots + a_n v_n + (-1)\, w = 0.$$

This is a nontrivial linear combination (the coefficient of w is $-1 \neq 0$) that equals zero, so $(v_1, \ldots, v_n, w)$ is linearly dependent. $\square$

Problem 4

(a)

A list (v) of length one is linearly independent if and only if $v \neq 0$.

Proof. By definition, (v) is linearly dependent if and only if there exists a scalar $a \in F$ with $a \neq 0$ such that $av = 0$.

($\Rightarrow$) Suppose $v \neq 0$. If $av = 0$ for some $a \in F$, then $a = 0$ (since $v \neq 0$ implies $a \cdot v = 0 \Rightarrow a = 0$ in any vector space). Hence the only solution to $av = 0$ is $a = 0$, so (v) is linearly independent.

($\Leftarrow$) Suppose $v = 0$. Then $1 \cdot v = 0$ with $1 \neq 0$, so (v) is linearly dependent, i.e. not linearly independent. □

(b)

A list (v_1, v_2) of length two is linearly independent if and only if neither vector is a scalar multiple of the other.

Proof. (v_1, v_2) is linearly *dependent* if and only if there exist $a_1, a_2 \in F$, not both zero, such that $a_1 v_1 + a_2 v_2 = 0$.

($\Rightarrow$) Suppose (v_1, v_2) is linearly dependent. Then $a_1 v_1 + a_2 v_2 = 0$ for some $(a_1, a_2) \neq (0, 0)$.

- If $a_1 \neq 0$, then $v_1 = -\frac{a_2}{a_1} v_2$, so v_1 is a scalar multiple of v_2.
- If $a_2 \neq 0$, then $v_2 = -\frac{a_1}{a_2} v_1$, so v_2 is a scalar multiple of v_1.

In either case at least one vector is a scalar multiple of the other. Equivalently, if neither is a scalar multiple of the other, then (v_1, v_2) is linearly independent.

($\Leftarrow$) Suppose $v_1 = \lambda v_2$ for some $\lambda \in F$ (the other case is symmetric). Then $1 \cdot v_1 + (-\lambda) \cdot v_2 = 0$ with $(1, -\lambda) \neq (0, 0)$, so (v_1, v_2) is linearly dependent, i.e. not linearly independent.

Combining both directions completes the proof. □

Problem 17

V is infinite-dimensional if and only if there exists a sequence $v_1, v_2, \ldots$ of vectors in V such that $(v_1, \ldots, v_m)$ is linearly independent for every positive integer m.

Proof. ($\Leftarrow$) Suppose such a sequence $v_1, v_2, \ldots$ exists. Assume for contradiction that V is finite-dimensional, say $\dim V = n$. Then $(v_1, \ldots, v_{n+1})$ is a linearly independent list of length $n+1$ in V. But any linearly independent list in an n-dimensional space has length at most n, a contradiction. Therefore V is infinite-dimensional.

($\Rightarrow$) Suppose V is infinite-dimensional. We construct the sequence inductively.

Base step. Since V is infinite-dimensional it is in particular non-trivial, so there exists $v_1 \in V$ with $v_1 \neq 0$. By Problem 4(a), (v_1) is linearly independent.

Inductive step. Suppose we have already found $v_1, \ldots, v_m$ such that $(v_1, \ldots, v_m)$ is linearly independent. Because V is infinite-dimensional, V cannot equal $\operatorname{span}(v_1, \ldots, v_m)$ (the span of a finite linearly independent list is a finite-dimensional subspace, whereas V

is infinite-dimensional). Hence there exists $v_{m+1} \in V \setminus \operatorname{span}(v_1, \ldots, v_m)$. By the linear dependence lemma (the contrapositive of Problem 3), since $v_{m+1} \notin \operatorname{span}(v_1, \ldots, v_m)$, the list $(v_1, \ldots, v_m, v_{m+1})$ is linearly independent.

By induction, we obtain a sequence $v_1, v_2, \ldots$ such that $(v_1, \ldots, v_m)$ is linearly independent for every positive integer m. □

Check in

I would like to brush my teeth after having my breakfast.